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PS C:\Users\ashak\Downloads\SingletonPatternExample\week1\Design Patterns and Principles\SingletonPatternExample\src> javac -d ../bin com/exercise1/singleton/*.java
PS C:\Users\ashak\Downloads\SingletonPatternExample\week1\Design Patterns and Principles\SingletonPatternExample\src> java -cp ../bin com.exercise1.singleton.SingletonTest
=== Test 1: Basic Singleton Check ===
[Logger] New instance created at 2026-06-22 18:12:29 by thread main
[2026-06-22 18:12:29] LOG: First message from logger1
[2026-06-22 18:12:29] LOG: Second message from logger2
logger1 hashCode: 1118140819
logger2 hashCode: 1118140819
logger1 == logger2 ? true
PASS: Only one instance exists.

=== Test 2: Multithreaded Singleton Check ===
[2026-06-22 18:12:29] LOG: Message from Worker-1
[2026-06-22 18:12:29] LOG: Message from Worker-4
[2026-06-22 18:12:29] LOG: Message from Worker-5
[2026-06-22 18:12:29] LOG: Message from Worker-6
[2026-06-22 18:12:29] LOG: Message from Worker-2
[2026-06-22 18:12:29] LOG: Message from Worker-0
[2026-06-22 18:12:29] LOG: Message from Worker-7
[2026-06-22 18:12:29] LOG: Message from Worker-3
[2026-06-22 18:12:29] LOG: Message from Worker-8
[2026-06-22 18:12:29] LOG: Message from Worker-9
All 10 thread instances identical? true
PASS: Singleton holds under concurrent access.
PS C:\Users\ashak\Downloads\SingletonPatternExample\week1\Design Patterns and Principles\SingletonPatternExample\src>
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PS C:\Users\ashak\Downloads\FactoryMethodPatternExample\FactoryMethodPatternExample> cd src
>> javac -d ../bin com/exercise2/documents/*.java
>> java -cp ../bin com.exercise2.documents.FactoryMethodTest
Opening Word document...
Saving Word document...
Is WordDocument? true

Opening PDF document...
Saving PDF document...
Is PdfDocument? true

Opening Excel document...
Saving Excel document...
Is ExcelDocument? true
PS C:\Users\ashak\Downloads\FactoryMethodPatternExample\FactoryMethodPatternExample\src>
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